



SG: China is heavily investing in compound semiconductors like GaN and silicon carbide, which are valuable for EVs and other power electronics applications. Given that compound semiconductors do not require advanced manufacturing technology, can China dominate this sector?

CM: Chinese companies and the Chinese government view compound semiconductors as a significant competitive advantage, with substantial investment driven by the growing EV market. China is likely to see increased protectionism and tariffs, which could provide guaranteed markets for Chinese firms, even if their quality does not match that of European and Japanese counterparts.

SG: China is focusing on lower-end chips like microcontrollers and Wi-Fi, producing them at lower costs using older technology. Could this lead to the US and others concentrating on high-end chips while China dominates the low-end market?

CM: Without intervention from policymakers in the US and other countries, China could indeed dominate the lower-end chip market. US, Japanese, and European leaders are concerned about China's subsidies impacting Western firms' profits and may impose more tariffs and restrictions to limit the presence of Chinese chips in global markets. This could result in complex geopolitical and regulatory battles.

SG: Companies such as Google, Amazon, and Microsoft are developing their own custom chips. Could this lead to a transformative shift in the industry similar to the fabless model of the 1980s?

CM: The key challenges for custom chips are cost and complexity. Currently, only major tech companies with substantial resources can afford

to develop them. To make custom chips more accessible to other industries, such as automotive and medical, we need cheaper, simpler design processes. AI-driven design software could help reduce costs and simplify development. Additionally, chiplet-based architectures, which involve assembling pre-designed components, could be an easier initial step for non-tech companies. However, further simplification and better design tools will be necessary for broader industry adoption.

SG: India introduced a comprehensive semiconductor policy around the same time your book was released. How do you assess the progress made in the past two and a half years?

CM: Considerable progress has been made under the India Semiconductor Mission. Major companies like Micron have started establishing facilities, and there have been announcements for OSAT and wafer fabrication facilities from Tata and others. Electronics product assembly in India is also expanding, with Apple planning to assemble 25% of its phones in India, contributing to the overall manufacturing ecosystem. Additionally, India's strength in chip design remains notable, and semiconductor product design should continue to be a key focus.

SG: Some believe India is not advancing as quickly as possible. Considering India started with minimal semiconductor manufacturing, what are your thoughts?

CM: Advancing in chip manufacturing and assembly is a challenging, capital-intensive process that requires several years to complete a single project. Given this context, progress has been solid. It is crucial not to rush the process but to develop a comprehensive ecosystem to support long-term growth.

SG: By 2030, is it realistic for India to have two semiconductor wafer fabs, two compound semiconductor fabs, and two OSAT plants?

CM: This is very realistic. It is certain that India will have at least two silicon fabs, although the specific types of semiconductors, technology nodes, and capacities will need to be clarified. The establishment of two OSAT/ATMP plants and two compound semiconductor fabs is also feasible.

SG: You often mention that large Capex manufacturing projects are like ribbon-cutting opportunities and that India should focus on design. What should be the focus and strategy for India to become a 'semiconductor product nation' rather than just a provider of design services?

CM: In the semiconductor industry, most financial gains accrue to companies producing semiconductor products, not merely providing design services. The challenge for India is to create brands and companies headquartered in India that sell globally. This has been achieved in the software sector, and there is no reason it cannot be replicated in semiconductors. India's venture capital ecosystem needs to become more hardware focused. Talent exists; it is about connecting it with business expertise and financing, and there are encouraging signs in this direction.

SG: The Indian diaspora has established many successful semiconductor companies in Silicon Valley. How can collaboration between Silicon Valley and India be improved to make India a 'semiconductor product nation'?

CM: Venture funding for hardware in India needs to increase and extend over longer periods, similar to Silicon Valley. While software ventures receive more VC attention, hardware projects require support